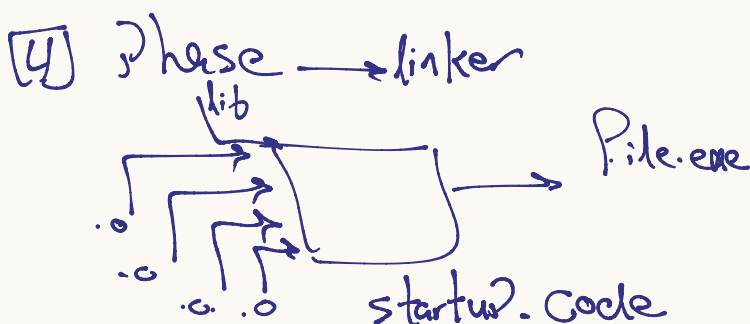
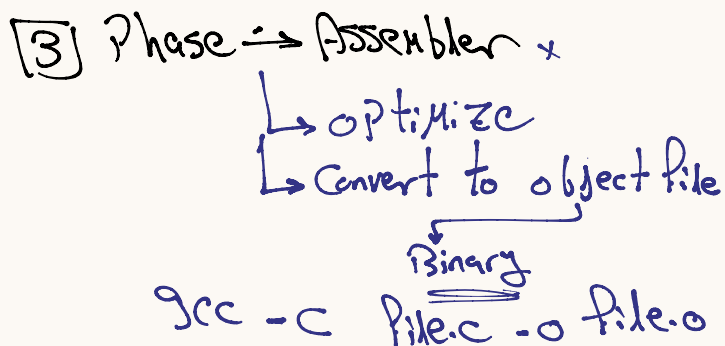
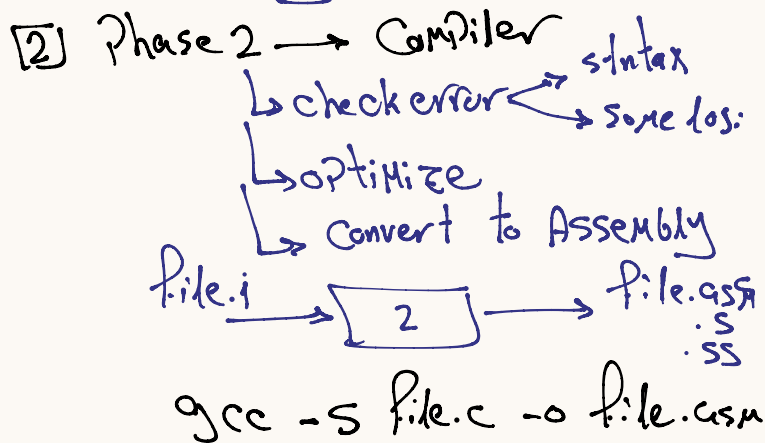
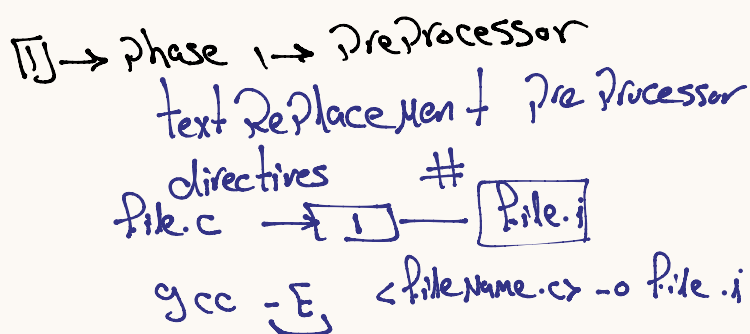




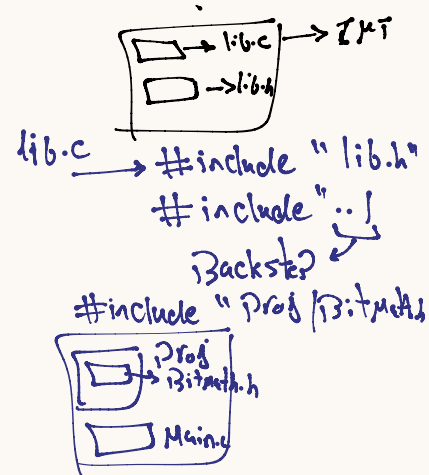
- ↳ text replacement
- ↳ directives

```
#include #define #undef  
#if #elif #else #endif  
#error #warning #ifdef #ifndef
```

- Pile must be Header Pile
- select Headertype

#include < > → system lib
#include " " → user define

Absolute → full path
relative → lib.c → I/O



```
#define ArraySize 5
uint8_t arr [ArraySize]
```

#define X 100 ←

===== X 100

#define X 200

===== X 200

$$\Rightarrow \# \text{ undef } x$$

→ #define function like Macro

Function

- Normal
- Macro ←
- Inline function

* Preprocessor Condition

* Configuration

#define version

config

2 →

#if version == 1

max speed = 100;

Air bag = 2;

Air condition = 20;

#elif version == 2

max speed = 200;

Air bag = 3;

Air condition = 18;

#else

#warning Invalid config

max speed = 200;

Air bag = 3;

Air condition = 18;

#endif

* Macro Function

#define Add(x, y)

(x+y)

int main()

↵ R = Add(5, 7); → 5+7

* Set Bit

#define set(Reg, n) Reg |= (1 << n)

int main()

↵ int x = 5;

// x |= (1 << 1b); ←

set(x, 1b);

Multi-line Function

#define HelloMessage() printf(1);
printf(1);
printf(1);

Macro Function

#define Add(x, y)

R = Add(5, 7) → 5+7
→ 50

Normal Function

Add → 4 Byte

Main → 50

#include "BitMath.h"

#include "BitMath.h"

config.h

↳ BitMath.h

config.h → x

BitMath.h

any header file

#ifndef x → File Name

#define x

#endif

BitMath.h →

#ifndef BITMATH_H

#define BITMATH_H